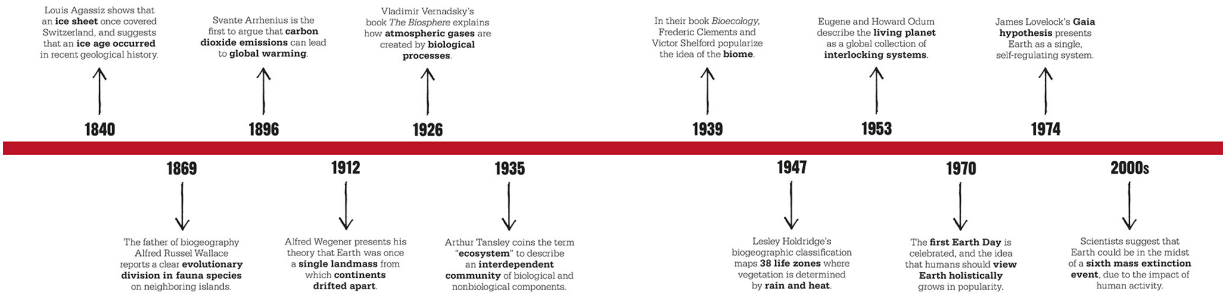
INTRODUCTION

For centuries, scientists in the Western world tried to reconcile the findings of geologists and fossil hunters with literal interpretations of biblical stories about Creation and the Great Flood. In 1654, for example, Archbishop Ussher dated Earth's creation to October 22, 4004 BCE. A series of discoveries challenged this narrative and led to new ideas about the dynamic history of life on Earth.



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Evidence in the rocks

Two Scottish geologists—James Hutton and Charles Lyell—advanced our understanding of Earth's age. In *Theory of the Earth* (1795), Hutton argued that the repeated cycles of sedimentation and erosion necessary to create thousands of feet of rock strata must indicate a truly ancient origin for the planet—an idea which Lyell developed further in the 1830s. Soon after, Swiss-American geologist Louis Agassiz proposed that the topography of some regions had been shaped by glaciations. Hutton and Lyell also noted that fossils of animals and plants vanished from the geological record. Lyell believed this to be evidence of extinction, challenging the prevailing belief that species were immutable.

Fossils also offered clues to movements of Earth's continents. German meteorologist Alfred Wegener noted that similar fossils could be found on